ELSEVIER

Brachytherapy 11 (2012) 224−229

BRACHYTHERAPY

The comparison of ruthenium brachytherapy and simultaneous transpupillary thermotherapy of choroidal melanoma with brachytherapy alone

Andrey A. Yarovoy*, Dzhavid A. Magaramov, Evgeniya S. Bulgakova

Ocular Oncology Department, The S. Fyodorov Eye Microsurgery Complex, Moscow, Russia

ABSTRACT

PURPOSE: To compare the outcomes of combined treatment of choroidal melanoma with ruthenium brachytherapy (BT) simultaneously with transpupillary thermotherapy (TTT) and treatment with BT alone.

METHODS AND MATERIALS: Two matched groups of patients, one treated with BT and simultaneous TTT (Group BT + TTT, $n = 63$), the other treated with BT alone (Group BT, $n = 70$) were analyzed retrospectively. The main outcome measures were rate of tumor regression, recurrences, enucleations, metastases, recurrence-free and overall survival rate, and visual acuity, assessed by Kaplan−Meier analysis.

RESULTS: Patients were matched according to mean age ($p = 0.22$), mean tumor thickness (6.4 vs. 6.25 mm, range 2.5−10.8 mm, $p = 0.59$), and mean length of followup (42 vs. 34.4 months, range 3−109, $p = 0.052$). Tumor largest basal diameter (13.0 vs. 12.9 mm), tumor location, and mean radiation dose (apical 135 vs. 136 Gy and scleral 1294 vs. 1438 Gy) were also similar in both groups ($p > 0.1$). Treatment with BT + TTT resulted in higher rate of tumor regression (63% vs. 49%, respectively, $p = 0.036$), lower 5-year tumor recurrence rate (96% vs. 83%, $p < 0.034$), and higher eye-globe preservation (98% vs. 87%, $p < 0.024$) and recurrence-free survival rates (89% vs. 67%, $p < 0.017$) than treatment with BT alone. There was no difference in complications ($p > 0.5$), metastasis-free (93% vs. 81%, $p > 0.22$) and overall survival rates (91% vs. 81%, $p > 0.39$), or in visual outcomes.

CONCLUSION: Combined treatment of choroidal melanoma with ruthenium BT and simultaneous TTT seems to provide higher local control, eye-globe preservation, and recurrence-free survival rates than treatment with BT alone and results in similar rates of metastases and overall survival.

Keywords: Brachytherapy; Choroidal melanoma; Transpupillary thermotherapy

Introduction

Episcleral plaque radiotherapy is one of the standard treatment options for uveal melanoma. The most common isotopes used for brachytherapy are ruthenium-106 (^{106}Ru) and iodine-125 with their well-known advantages and disadvantages. The most important limitation of ^{106}Ru brachytherapy (BT) is the suitable tumor thickness, generally considered to be about 5−7 mm (1−4), although there are some studies of ^{106}Ru BT of tumors with thickness up to 11.8 mm (5). The proposal to use BT in combination with a new method of treatment transpupillary thermotherapy (TTT) (6) was associated with the hope, among other things, to make it possible to treat thicker tumors and improve the globe salvage rate (7). By contrast with the dozens of existing studies of TTT as a primary treatment of choroidal melanoma, including those covering the largest groups of treated patients (8−12), only a few reports on the combined treatment of choroidal melanoma with BT and TTT have been published to date (13−18). These studies described nonuniform patterns of application of TTT with BT (just simultaneously (7, 17), simultaneously and with additional TTT during the followup (15), after

Received 7 April 2011; received in revised form 15 August 2011; accepted 8 September 2011.

* Corresponding author. Ocular Oncology Department, The S. Fyodorov Eye Microsurgery Complex, 59A, Beskudnikovsky Blvd, Moscow 127486, Russia. Tel.: +7-495-488-89-43; fax: +7-495-485-59-54.

E-mail address: yarovoyaa@yandex.ru (A.A. Yarovoy).

1538-4721/$ - see front matter
doi:10.1016/j.brachy.2011.09.007

BT for tumor recurrences (6), at the end of BT, and within several months after BT (14, 16), etc.), various isotopes were used (14), most studies had limited number of reported cases. Moreover, there are few comparative studies of the combined treatment and BT as monotherapy (18, 19). Hence the real benefits of additional TTT are not quite clear. In this report, we present the findings resulting from our comparative study of the outcomes of the combined treatment of choroidal melanoma with ^{106}Ru BT and simultaneous TTT as well as those of BT alone.

Methods and materials

The charts of 520 patients with choroidal melanoma who were treated with ^{106}Ru BT with or without simultaneous TTT at the Ocular Oncology Department of the S. Fyodorov Eye Microsurgery Complex, Moscow, Russia, between November 1999 and December 2008, were reviewed retrospectively. The analysis started in November 2009. This study was approved by the Institutional scientific committee of The S. Fyodorov Eye Microsurgery Complex, Moscow, Russia. Written informed consent before treatment had been obtained from each patient. Patients who underwent BT in combination with TTT were classified as Group BT + TTT, whereas patients who underwent BT alone were classified as Group BT.

Only patients with tumor thickness over 2.5 mm, without iris involvement, with a followup longer than 12 months in the cases of positive results, without other types of cancer, and those with all available necessary information about clinical parameters were included in this study. Positive results were defined as full (complete) or incomplete (partial) tumor regression without continuous growth or recurrence, absence of enucleation, metastasis, or death. Patients with bilateral tumors and those treated with TTT as an additional treatment of residual tumor after BT were not included in this study. Patients with tumor thickness over 10 mm were treated with BT + TTT or with BT because they declined primary enucleation. None of the patients had metastases at the time of diagnosis and treatment.

BT was performed using ^{106}Ru + ^{106}Rh plagues with safety margins of 1.5–2 mm. For BT, a standard set of ^{106}Ru ophthalmic plagues of different diameter (active part: 14, 16, and 19 + 2.5 mm of inactive rim) and shape (round and notched) was used (manufacturer—State scientific center of the Russian Federation—Institute for physics and power engineering named after A.I. Leypunsky, Obninsk, Russia). Radius of the inner surface of the plague is 12 mm, thickness is 1 mm, nominal nuclide activity of different plaques is from 0.779 to 1.907 mCi, mean dose rates at the center of the plague concave side is from 1070 to 2005 cGy/h. Standard deviations of dose rate on the radioactive surface of the plagues is from 4.1% to 15%. The dose prescription point was tumor height plus 1 mm for the sclera. Standard prescription dose to the tumor apex was 120–160 Gy. The range and mean levels of prescribed doses are shown in Table 1. In the cases of tumors higher than 6.0 mm, the dose prescription point was sclera with the radiation dose not exceeding 2700 Gy. TTT simultaneously with BT was performed in the cases of possibility to put the contact lens on the ocular surface, in the cases of sufficient transparency of ocular media, absence of retinal detachment, or hemorrhages on the tumor surface. TTT was performed with an 810-nm diode laser within 24–48 hours of the plague fixation. A laser irradiation (2.5–3.0 mm beam; laser power of 550–850 mW) was applied to the maximum area of the tumor surface.

All patients were followed up every 3 months within the first year, every 6 months within the 2–5 years after the treatment, and yearly subsequently. The followup included ocular examination and ocular echographic examination (standardized A- and B-scans; Ultrascan, Alcon, Fort Worth, USA) (20) and metastatic screening.

Collected data included patient's age, gender, pre- and posttreatment tumor dimensions, location, visual acuity before and after the treatment, apical and scleral radiation doses, tumor regression rate, complications, length of followup, outcomes, and additional treatment of failures. The main outcome measures were tumor recurrences, enucleations, metastases, recurrence-free and overall survival, and best-corrected visual acuity.

Complete tumor regression was defined as a regression to a stable ophthalmoscopic and sonographic scar without or with residual thickness not exceeding 2.5 mm (with one exception, enucleated eye with secondary glaucoma and 5.6 mm totally necrotic tumor after BT + TTT) and without signs of viable tumor tissue. Incomplete (partial) regression was defined as a regression of any degree to residual tumor of some thickness with ophthalmoscopic signs of possible presence of viable tumor with stabilization of its size during the followup. Recurrence was defined as signs of an increase in residual tumor size (1.0 mm or more) after the initial regression within the followup. Recurrence-free rate was calculated as the proportion of patients without immediate failures of treatment and recurrences within the followup. Recurrence-free survival rate was calculated as the proportion of patients without immediate treatment failures, recurrences, and metastases.

The data were analyzed with Statistica software (Stat Soft, Inc., Tulsa, OK, USA). Descriptive statistics were given as the mean and range. Continuous variables were compared using the Wilcoxon matched pairs test, and Yates-corrected chi-square test was used for the comparison of frequencies. Kaplan–Meier estimates with comparison by log-rank test were used to analyze the main outcome measures as a function of time. The chosen level of statistical significance was $p < 0.05$.

Results

Two groups of patients were matched according to age, tumor thickness, and length of followup (mean and median values were similar in both groups (Table 1). Final

Table 1
Comparison of the baseline characteristics of 133 matched patients who underwent ^{106}Ru brachytherapy simultaneously with transpupillary thermotherapy (BT + TTT) and ^{106}Ru brachytherapy as a monotherapy (BT)

Characteristics	BT + TTT ($n = 63$)	BT ($n = 70$)	p[a]
Matching variables			
Age (y)			
Mean	50.7 ± 11.3	54.5 ± 10.9	0.22[b]
Median	51	53.5	
Range	26–71	31–74	
Tumor thickness (mm)			
Mean	6.4 ± 2.0	6.25 ± 1.7	0.59[b]
Median	6.3	6.3	
Range	2.5–10.8	3.5–10.5	
Followup (months)			
Mean	42 ± 22.4	34.4 ± 22.8	0.052[b]
Median	39	29.5	
Range	6–84	3–109	
Other baseline variables			
Male/female	30/33	31/39	0.83[c]
Mean largest basal diameter, mm (range)	13.0 ± 2.3 (6.8–18.9)	12.9 ± 2.4 (6.0–18.0)	0.53[b]
Number of patients in subgroups with tumor thickness ≤6.0/>6.0 mm	27/36	33/37	0.75[c]
Mean tumor thickness in subgroup ≤6.0 mm	4.6 ± 0.17	4.75 ± 0.65	0.29[b]
Mean tumor thickness in subgroup >6.0 mm	7.8 ± 0.29	7.6 ± 1.13	0.57[b]
Number of juxtapapillar tumors	10	5	0.19[c]
Number of ciliochoroidal tumors	2	6	0.35[c]
Mean apical radiation dose, Gy (range)	135 ± 31 (90–190)	136 ± 20 (108–179)	0.32[b]
Mean scleral radiation dose, Gy (range)	1294 ± 761 (295–2584)	1438 ± 774 (512–2676)	0.73[b]

BT = brachytherapy; TTT = transpupillary thermotherapy.
[a] BT + TTT compared with BT.
[b] Wilcoxon matched pairs test.
[c] Yates-corrected chi-square test.

comparative analysis included 133 patients treated either with BT + TTT ($n = 63$) or with BT alone ($n = 70$). Sixty-one (97%) brachytherapy procedures in the BT + TTT group and 69 (99%) brachytherapy procedures in the BT group were performed by a single surgeon (AAY). The two groups were comparable with respect to main variables (Table 1).

In the BT + TTT group, there was 1 patient (tumor thickness: 9.7 mm) who did not respond to the treatment (additional BT was performed), and in the BT group, there were 3 patients (tumor thickness: 7.1, 7.9, and 8.8 mm) of immediate treatment failures with tumor progression within 3–4 months after treatment and subsequent enucleations. Mean regression rate (for those patients who had any regression) in the BT + TTT group was 63%, from 6.42 ± 0.24 mm to 2.4 ± 1.91 mm, and in the BT group, the mean regression rate was 49%, from 6.17 ± 1.9 mm to 3.14 ± 1.9 mm ($p = 0.036$). Complete tumor regression was achieved in 52% of patients after BT + TTT and in 29% after BT ($p = 0.009$). There was no difference (61% and 54%, respectively) in regression rates between the ≤6.0-mm subgroups of the two groups ($p = 0.296$), but in >6.0-mm subgroups, the difference (63% and 47%) was significant ($p < 0.019$).

Tumor recurrences occurred in one case in the BT + TTT group (30 months after the treatment of a 6.1-mm tumor), which was controlled with additional BT, and in 6 cases in the BT group (mean: 20.5 months after treatment; range, 8–43 months, mean pretreatment tumor thickness, 7.4 mm; range, 4.7–9.9 mm) that were managed with additional BT ($n = 3$) or enucleation ($n = 3$). According to Kaplan–Meier estimates, 5-year recurrence-free rate was higher ($p < 0.034$) after combined treatment (96%) than after BT alone (83%) (Fig. 1a).

One patient in the BT + TTT group required enucleation because of the painful secondary glaucoma with a 5.6-mm totally necrotic tumor. In the BT group, 8 patients underwent enucleation: 3 patients because of the absence of treatment effect, 2 patients because of the tumor recurrence (5.5 and 7.4 mm), 2 patients because of uncontrolled neovascular glaucoma (6.5 and 7.9 mm), and 1 patient because of scleral melt (7.4 mm). BT + TTT resulted in a higher eye-retention rate (98%) than BT alone (87%) ($p < 0.024$) (Fig. 1b). The same was true for the >6.0-mm subgroup ($p = 0.036$). It was not possible to estimate the enucleation rate in the ≤6.0-mm subgroup as there was only one enucleation patient in the BT group.

Three patients in the BT + TTT group (18, 30, and 32 months after treatment), and 6 patients in the BT group (mean followup, 31.7 months; range, 16–49 months) developed metastases. One patient in the BT + TTT group died of an unrelated cause (gastric cancer).

There was practically no difference between the two groups with respect to the 5-year metastasis-free rate,

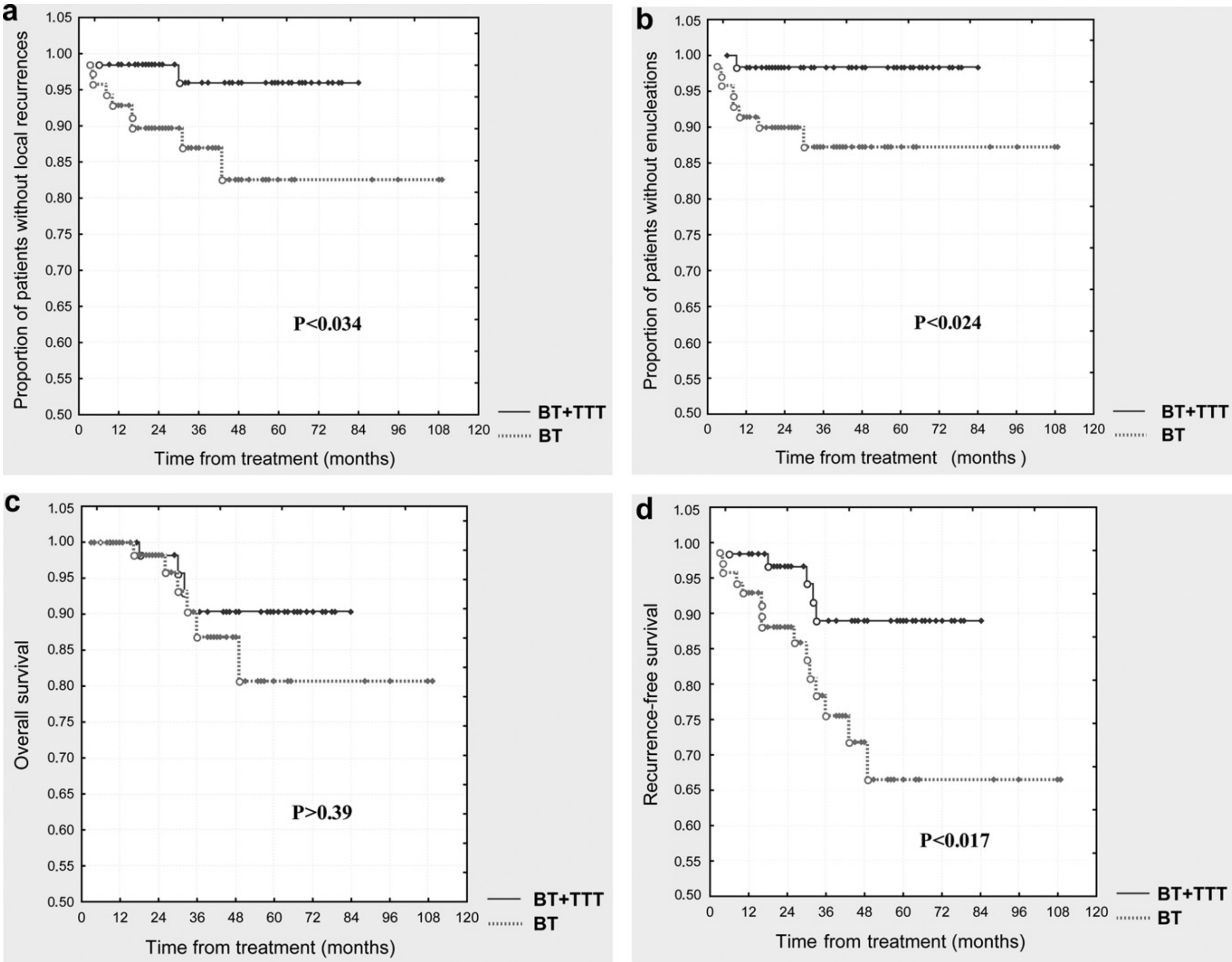

Fig. 1. (a) Kaplan–Meier estimates of proportion of patients free of local recurrences, (b) free of enucleations, (c) of the rates of overall survival, (d) and of recurrence-free survival over time in patients treated with brachytherapy simultaneously with transpupillary thermotherapy (BT + TTT, $n = 63$) or with brachytherapy as a sole treatment (BT, $n = 70$). BT = brachytherapy; TTT = transpupillary thermotherapy.

93% and 81%, respectively ($p > 0.22$). The same was true for the ≤6.0-mm subgroup ($p > 0.1$) and the >6.0-mm subgroup ($p > 0.63$). Overall survival after BT + TTT and BT was similar, 91% and 81%, respectively ($p > 0.39$) (Fig 1c). Five-year recurrence-free survival rate was significantly higher ($p < 0.017$) after BT + TTT (89%) than after BT alone (67%) (Fig. 1d).

The total amount of complications was 61 after BT + TTT and 67 after BT. In the BT + TTT group, 36 patients (57%) had at least one kind of complication and in the BT group, 44 patients (63%). The BT + TTT and BT groups had similar number of complications in the observation period (the amount of complications in the groups was calculated for patients without immediate treatment failures): retinopathy, 34% ($n = 21$) and 33% ($n = 22$); partial optic nerve atrophy, 16% ($n = 10$) and 19% ($n = 13$); optic neuropathy 13% ($n = 8$) and 5% ($n = 3$); retinal detachment, 11% ($n = 7$) and 10% ($n = 7$); optic nerve atrophy, 6% ($n = 4$) and 9% ($n = 6$); vitreous hemorrhage, 6% ($n = 4$) and 2% ($n = 1$); maculopathy, 5% ($n = 3$) and 8% ($n = 5$); cataract, 5% ($n = 3$) and 3% ($n = 2$); secondary glaucoma, 2% ($n = 1$) and 6% ($n = 4$); scleral melting, 0% ($n = 0$) and 5% ($n = 3$); and phthisis bulbi, 0% ($n = 0$) and 2% ($n = 1$), respectively.

The posttreatment best-corrected visual acuity was analyzed only in patients without enucleations or repeated BT as it was measured at the end of the followup. The significant level of visual acuity change was considered to be significant at 0.1 or higher. No differences were found between the two groups in visual outcomes. Seven percent of patients ($n = 4$) in both groups increased their best-corrected visual acuity, 22% ($n = 13$) in the BT + TTT group and 29% ($n = 17$) in the BT group did not change, and 71% ($n = 43$) and 64% ($n = 38$) decreased their visual acuity ($p > 0.5$). Both treatments caused similar changes in the number of patients in subgroups according to visual acuity. The number of patients with visual acuity of 1.0–0.5 decreased by 38% (from 29 to 6 patients) in the

BT + TTT group and by 41% (from 28 to 4 patients) in the BT group. The number of patients with visual acuity of 0.4–0.2 did not change ($n = 11$) after BT + TTT and decreased by 8% (from 12 to 7 patients) after BT. The number of patients with visual acuity of 0.1 and worse increased in both groups by 39% (from 20 to 43 patients) and by 49% (from 19 to 48 patients), respectively. No statistical difference between the both groups before ($p = 0.96$) and after treatment ($p = 0.459$) was found.

Discussion

The presumed advantages of combined treatment of uveal melanoma with ^{106}Ru BT and TTT (6, 7) needed to be estimated in comparison with BT as monotherapy. Although previous reports did not show any benefits of additional TTT (19), the present study has revealed some significant advantages of BT + TTT over BT alone. To our knowledge, this study, comparing the two methods, covers the largest groups of treated patients. Both groups were comparable with respect to the main tumor characteristics, radiation doses, and followup. Although the followup after BT was insignificantly shorter than after BT + TTT, this fact does not alter the validity of comparison but can emphasize the advantages of combined treatment.

According to our data, tumor regression was more complete after simultaneous use of TTT and BT especially in tumors over 6.0- mm thick, including a larger number of cases with complete regression than after BT alone. A similar mean regression rate after combined treatment in a comparable group of patients was reported by Kreusel et al. (16). More complete tumor regression after BT + TTT led to fewer recurrences and a higher 5-year recurrence-free rate (96%) than after BT alone. This rate is higher or similar to the previously reported local results after ^{106}Ru BT as a sole treatment (1, 3, 21–24), but the majority of these studies included much smaller tumors.

The higher local control rate after BT + TTT resulted in the higher 5-year eye-retention rate (98%) in comparison with eye-retention rate after BT in the present study and with eye-retention rate after BT in other studies (1–4, 21–24). The rate of enucleations after BT + TTT in our study was similar to other published results of combined treatment for choroidal melanoma (7, 14–17). In 4 of 8 patients, enucleations were conducted after BT because of the unsuccessful local control of tumors >6.0 mm in thickness and in 3 patients because of complications. In the compared groups, the rates and spectrum of complications did not differ significantly and were higher than those in other reports on ^{106}Ru BT (1, 2, 4, 22, 24, 25). We used somewhat unconventionally high irradiation doses, up to 2700 Gy to the sclera, but our experience has showed that it is possible to prescribe such high doses without fatal rising of the complication level. Hence the visual outcomes after both treatments were similar and not high. All complications were the result of radiation side effects but not of TTT.

Numerous reports have shown that various methods of treatment of choroidal melanoma of the same size had similar rates of metastases (26–31). In our study, 5-year metastatic and overall survival rates after BT with or without additional TTT did not differ and corresponded to the previously published data. Because of higher local outcomes of BT + TTT, the combined treatment resulted in 89% of recurrence-free survival that was higher than after BT alone (67%). The number of metastases was slightly lower after BT + TTT but of no statistical significance. A lower tumor recurrence rate may result in a lower metastatic risk as was found in prior reports (32).

When comparing the outcomes of BT with TTT reported in other studies, the difference in the pattern of combining these methods, mentioned above, should be kept in mind. According to our data, it seems that simultaneous use of BT and TTT can make it possible to treat choroidal melanoma with a thickness higher than 5–6 mm with ^{106}Ru plague radiotherapy and improve its' local and eye-preserving outcomes. Our study was of a retrospective nature, and for more thorough conclusions, prospective randomization of patients is essential.

Conclusion

Combined treatment of choroidal melanoma with ruthenium BT and simultaneous TTT seems to provide higher local control, eye-globe preservation, and recurrence-free survival rates than treatment with BT alone and results in similar rates of metastases and overall survival. Our study was of a retrospective nature, and for more thorough conclusions prospective randomization of patients is essential.

References

[1] Lommatzsch PK. Results after β–irradiation (Ru–106/Rh–106) of choroidal melanomas: 20 years experience. *Br J Ophthalmol* 1986;70:844–851.

[2] Rouberol F, Roy P, Kodjikian L, *et al*. Survival, anatomic, and functional long-term results in choroidal and ciliary body melanoma after ruthenium brachytherapy (15 years' experience with beta-rays). *Am J Ophthalmol* 2004;137:893–900.

[3] Bergman L, Nilsson B, Lundell G, *et al*. Ruthenium brachytherapy for uveal melanoma, 1979-2003: Survival and functional outcomes in the Swedish population. *Ophthalmology* 2005;112:834–840.

[4] Heindl LM, Lotter M, Strnad V, *et al*. High-dose 106Ruthenium plaque brachytherapy for posterior uveal melanoma. A clinicopathologic study. *Ophthalmologe* 2007;104:149–157.

[5] Kaiserman I, Anteby I, Chowers I, *et al*. Post-brachytherapy initial tumour regression rate correlates with metastatic spread in posterior uveal melanoma. *Br J Ophthalmol* 2004;88:892–895.

[6] Journee-de Korver JG, Oosterhuis JA, De Wolff-Rouendaal D, *et al*. Histopathological findings in human choroidal melanomas after transpupillary thermotherapy. *Br J Ophthalmol* 1997;81:234–239.

[7] Seregard S, Landau I. Transpupillary thermotherapy as an adjunct to ruthenium plaque radiotherapy for choroidal melanoma. *Acta Ophthalmol Scand* 2001;79:19–22.

[8] Shields CL, Shields JA, Peres N, *et al*. Primary transpupillary thermotherapy for small choroidal melanoma in 256 consecutive cases: Outcomes and limitations. *Ophthalmology* 2002;109:225–234.

[9] Aaberg TM, Bergstrom CS, Hickner ZJ, *et al*. Long-term results of primary transpupillary thermal therapy for the treatment of choroidal malignant melanoma. *Br J Ophthalmol* 2008;92:741–746.

[10] Yarovoy AA, Magaramov DA, Bulgakova ES. Which choroidal melanoma should be treated with primary transpupillary thermotherapy? Our experience from 78 patients. *Eur J Ophthalmol* 2010;20: 186–193.

[11] Parrozzani R, Boccassini B, De Belvis V, *et al*. Long-term outcome of transpupillary thermotherapy as primary treatment of selected choroidal melanoma. *Acta Ophthalmol* 2009;87:789–792.

[12] De Potter P, Jamart J. Adjuvant indocyanine green in transpupillary thermotherapy for treatment for choroidal melanoma. *Ophthalmology* 2003;110:406–414.

[13] Oosterhuis JA, Journee-de Korver JG, Keunen JE. Transpupillary thermotherapy: Results in 50 patients with choroidal melanoma. *Arch Ophthalmol* 1998;116:157–162.

[14] Shields CL, Cater J, Shields JA, *et al*. Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: Tumor control and treatment complication in 270 consecutive patients. *Arch Ophthalmol* 2002;120:933–940.

[15] Bartlema YM, Oosterhuis JA, Journee-De Korver JG, *et al*. Combined radiotherapy and transpupillary thermotherapy in choroidal melanoma: 5 years' experience. *Br J Ophthalmol* 2003;87:1370–1373.

[16] Kreusel KM, Bechrakis N, Riese J, *et al*. Combined brachytherapy and transpupillary thermotherapy for large choroidal melanoma: Tumor regression and early complications. *Graefes Arch Clin Exp Ophthalmol* 2006;244:1575–1580.

[17] Stoffelns BM, Kutzner J, Schöpfer K, *et al*. Prospective nonrandomized analysis of ''Sandwich Therapy'' for malignant melanoma of the choroid. *Klin Monbl Augenheilkd* 2002;219:211–215.

[18] Stoffelns BM, Schöpfer K, Jochem T, *et al*. Tumor regression in malignant choroidal melanomas after transpupillary thermotherapy (TTT) versus ruthenium brachytherapy and sandwich therapy—A comparative analysis. *Klin Monbl Augenheilkd* 2010;227:262–268.

[19] Gündüz K, Kurt RA, Akmeşe HE, *et al*. Ruthenium-106 plaque radiotherapy alone or in combination with transpupillary thermotherapy in the management of choroidal melanoma. *Jpn J Ophthalmol* 2010;54:338–343.

[20] Ossoinig KC. Standardized echography: Basic principles, clinical applications, and results. *Int Ophthalmol Clin* 1979;19:127–210.

[21] Damato B, Patel I, Campbell IR, *et al*. Local tumor control after 106Ru brachytherapy of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:385–391.

[22] Isager P, Ehlers N, Urbak SF, *et al*. Visual outcome, local tumour control, and eye preservation after 106Ru/Rh-106 brachytherapy for choroidal melanoma. *Acta Oncol* 2006;45:285–293.

[23] Stoffelns BM, Kutzner J, Jochem T. Retrospective analysis of ruthenium-106 brachytherapy for small and medium-sized malignant melanoma of the posterior choroid. *Klin Monbl Augenheilkd* 2002; 219:216–220.

[24] Hermann RM, Pradier O, Lauritzen K, *et al*. Does escalation of the apical dose change treatment outcome in beta–irradiation of posterior choroidal melanomas with Ru–106 plaques? *Int J Radiat Oncol Biol Phys* 2002;52:1360–1366.

[25] Shields CL, Naseripour M, Cater J, *et al*. Plaque radiotherapy for large posterior uveal melanomas (> or =8 mm thick) in 354 consecutive patients. *Ophthalmology* 2002;109:1838–1849.

[26] Foulds WS, Damato BE, Burton RL. Local resection versus enucleation in the management of choroidal melanoma. *Eye (Lond)* 1987; 1:676–679.

[27] Augsburger JJ, Correa ZM, Freire J, *et al*. Long-term survival in choroidal and ciliary body melanoma after enucleation versus plaque radiation therapy. *Ophthalmology* 1998;105:1670–1678.

[28] Seregard S. Long–term survival after ruthenium plaque radiotherapy for uveal melanoma. A meta–analysis of studies including 1066 patients. *Acta Ophthalmol Scand* 1999;77:414–417.

[29] Wilson MW, Hungerford JL. Comparison of episcleral plaque and proton beam radiation therapy for the treatment of choroidal melanoma. *Ophthalmology* 1999;106:1579–1587.

[30] Cohen VM, Carter MJ, Kemeny A, *et al*. Metastasis-free survival following treatment for uveal melanoma with either stereotactic radiosurgery or enucleation. *Acta Ophthalmol Scand* 2003;81: 383–388.

[31] Gambrelle J, Grange JD, Devouassoux Shisheboran M, *et al*. Survival after primary enucleation for choroidal melanoma: Changes induced by the introduction of conservative therapies. *Graefes Arch Clin Exp Ophthalmol* 2007;245:657–663.

[32] Harbour JW, Char DH, Kroll S, *et al*. Metastatic risk for distinct patterns of postirradiation local recurrence of posterior uveal melanoma. *Ophthalmology* 1997;104:1785–1792. discussion 1792–1793.